

Reinforcement Learning and Agentic Optimization for Quantum Gates and Quantum Sensing with Electrons on Helium

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Reinforcement learning (RL) already optimizes the double-well trap potential for entanglement-based quantum sensing on the electrons-on-helium (eHe) platform, using a proximal-policy-optimization (PPO) agent maximizing a Fisher-information-based reward [?]. High-fidelity two-qubit gates on the same platform, by contrast, are currently designed by grid search over a low-dimensional pulse-shape family [?]. We propose extending RL-based optimization to gate design, and further explore a broader “agentic” workflow in which higher-level control (hyperparameter choice, reward shaping, campaign design across the gate/sensing/readout stack) is itself automated or human-AI collaborative. This is a proposal manuscript: we outline the RL environment for gate optimization, a joint gate-and-sensing reward that could unify the two existing use cases, and the open questions that need to be resolved before results can be reported.

I. INTRODUCTION

Model-free reinforcement learning has shown broad success in quantum control tasks — from generic state-preparation and gate-design problems [? ?] to Bayesian-optimal experimental design in quantum metrology [?] — and is reviewed extensively for quantum technology applications in general [?]. On the electrons-on-helium platform specifically, an Actor-Critic PPO agent [?] already optimizes the trap electrode voltages to maximize a composite reward built from singlet/triplet state fidelities and classical Fisher information, for the purpose of entanglement-enhanced sensing [?]. The two-qubit gate work on the same platform [?], in contrast, optimizes a much smaller parameter set (ramp and hold times of an erf-shaped voltage pulse) by grid search — a choice that was reasonable given the low dimensionality of that particular pulse family, but leaves open whether a richer, RL-discoverable pulse shape could improve gate fidelity, robustness, or speed beyond the current best results ($F = 0.999$ for $\sqrt{\text{iSWAP}}$, $F = 0.996$ for CZ).

This project has two parts. First, the more concrete one: port the existing RL/PPO infrastructure from the sensing use case to gate design, replacing the current grid search over $(t_{\text{ramp}}, t_{\text{hold}})$ with a policy that controls the full time-dependent voltage trajectory $V(\lambda(t))$, and see whether it finds higher-fidelity or more robust gates than the erf-ramp family currently assumed. Second, the more exploratory one: since the same underlying device (trap electrodes, Hartree/CI machinery, resonator readout) supports both gates and sensing, we ask whether a single agentic framework — potentially a hierarchical or multi-objective one — can jointly optimize across both tasks, or even autonomously propose which numerical experiments to run next (an “agentic AI” layer on top of classical RL, in the sense of an agent that plans and sequences its own investigation rather than only optimizing a fixed reward on a fixed task).

II. MODEL

A. RL environment for gate design

Reusing the two-qubit gate model of Ref. [?]: state $s_t = (\text{present voltage vector } V(t) \text{ or its parametrization } \lambda(t), \text{ elapsed pulse time, present gate-state overlaps } U_{ij}(t))$; action $a_t = \text{increment to the voltage trajectory (either a direct increment to } V(t + \delta t), \text{ or coefficients of a smooth basis such as a truncated Fourier/spline expansion of } \lambda(t), \text{ to keep the action space low-dimensional and the resulting pulses physically smooth)}$; reward = a combination of final gate fidelity F (Sec. IV of Ref. [?]) and a speed penalty, evaluated via the same Crank-Nicolson propagation already used for the grid-search approach. Candidate architectures: PPO (consistent with the existing sensing-RL infrastructure [?]), or a demonstration-seeded approach that starts from the known erf-ramp solution and lets RL refine it (in the spirit of reinforcement-learning-from-demonstration approaches to quantum control) rather than learning a pulse shape from scratch.

B. A joint gate-and-sensing reward (exploratory)

Because both tasks share the same trap/CI backbone, a natural extension is a multi-objective reward

$$R(a) = w_{\text{gate}} R_{\text{gate}}(a) + w_{\text{sense}} R_{\text{sense}}(a), \quad (1)$$

where R_{sense} is the existing Fisher-information-based reward of Ref. [?] and R_{gate} is the fidelity/speed reward above, with weights $w_{\text{gate}}, w_{\text{sense}}$ set by the application (a device optimized purely for sensing need not also be a good gate, and vice versa — the interesting question is how much is given up by demanding both). This would let a single agent characterize the trade-off surface between “good qubit” and “good sensor” trap configurations, rather than optimizing each in isolation as is currently done.

C. Agentic layer (exploratory, least defined)

Beyond parameter optimization within a fixed RL environment, an “agentic” framing would have a higher-level controller: choose which reward to optimize, when to switch tasks, when a numerical result warrants a follow-up simulation, or when to flag a result as inconsistent with the established claims-discipline conventions of this project (see `skills/literature-and-notation`). This is the least concrete part of the proposal and needs scoping before any implementation is attempted — see open questions below.

D. Open questions

- Does an RL-discovered pulse shape actually beat the erf-ramp family of Ref. [?], or does the existing grid search already saturate what’s achievable for this Hamiltonian? (Plausible null result — state it as such if so.)
- What is the right action-space parametrization to keep training sample-efficient while still letting the agent discover genuinely new pulse shapes (not just re-derive the erf ramp)?
- Is the joint gate-and-sensing reward actually useful, or does it just produce a mediocre compromise with no practical use case? Needs a concrete motivating

scenario (e.g. a device that must do both without reconfiguration) before investing in it.

- What, concretely, would distinguish an “agentic” layer from a slightly more elaborate hyperparameter search? This needs a sharper definition before it’s a research claim rather than a buzzword — see Ref. [?] for how the broader RL-for-quantum-technology literature currently scopes this kind of question.

III. RESULTS

Not yet available.

IV. DISCUSSION AND OUTLOOK

If the gate-RL part succeeds, the natural next step is a systematic comparison against the grid search of Ref. [?] across the same fidelity/robustness metrics (swap error, leakage error, sensitivity to ramp/hold-time deviations), reusing the numerical toolkit in `ci-dvr-hartree-numeric`s. The joint reward and agentic-layer parts should be scoped down to a single, concrete, falsifiable claim before further development.

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Placeholder.

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